EL-HOUCINE OUSSI

Software Engineer • Machine Learning Engineer

oussi.elhoucine0@gmail.com | +86 177 5100 8739 (CN) / +212 642 587 839 (MA) | Nanjing, China & Oujda, Morocco

oussi108.github.io | github.com/Oussi108

Nationality: Moroccan | **Current Status:** Full-time Master's Student at NUIST (Expected Graduation: June 2027) | **Availability:** Open to Global Relocation & Remote Roles

TECHNICAL COMPETENCIES

Languages: Python, C# (.NET), Dart, Java, C/C++, TypeScript, JavaScript, SQL, Bash / Shell

Frameworks & ML: ASP.NET Core 8 / MVC, Flutter, PyTorch, LightGBM, XGBoost, Scikit-learn, Django, Spring Boot, React

Cloud & Databases: PostgreSQL, SQL Server, MySQL, SQLite, MongoDB, Firebase (Auth/Firestore), Docker, Git, Linux, IIS

System Architecture: Clean Architecture, RESTful APIs, MVVM (Riverpod), Microservices, Spatio-Temporal Time Series, CI/CD

WORK & RESEARCH EXPERIENCE

NUIST Renewable Energy AI Lab — Graduate AI & Data Science Researcher — Sep 2024 – Present | Nanjing, China

- **Machine Learning Benchmarking:** Engineered scalable benchmark pipelines across 50 commercial Australian solar farms comparing persistence, GBDT, and deep neural models for multi-horizon power forecasting.
- **Spatio-Temporal Data Engineering:** Processed multi-terabyte environmental datasets, ingesting and aligning Himawari-8/9 geostationary satellite imagery with ECMWF ERA5 numerical weather fields.
- **Scientific Reproducibility:** Published open-source benchmark dataset & evaluation suite on Zenodo (DOI: 10.5281/zenodo.23590205) with 100% reproducible execution recipes and checksum manifests.

Majal Berkane – Smart City — Backend & Mobile Developer Intern — Mar 2024 – Jul 2024 | Berkane, Morocco

- **Mobile Architecture (Flutter):** Designed responsive cross-platform application adopting MVVM architecture with Riverpod for robust and decoupled state management.
- **Backend Systems (.NET Core):** Engineered high-throughput RESTful Web APIs with ASP.NET Core following Clean Architecture, CQRS, and repository patterns.
- **Database & IoT Optimization:** Tuned SQL Server queries, indexing, and in-memory caching for live IoT sensor streams, cutting data retrieval latency by over 35%.

Direction Provinciale de l'Équipement, du Transport et de la Logistique — Web Engineering Intern Mar 2023 – Jul 2023 | Oujda, Morocco

- **Full-Stack Web Development:** Built internal administrative platform in Django automating personnel records, heavy equipment dispatch, and logistics approval workflows.
- **Data Analytics Automation:** Implemented role-based access control (RBAC) and automated reporting pipelines using Pandas and NumPy for operational metrics.
- **Production Deployment:** Deployed application on Windows Server with IIS reverse proxy configuration, security hardening, and automated SQL Server database backups.

EDUCATION

Nanjing University of Information Science and Technology (NUIST) — Nanjing, China — Sep 2024 – Jun 2027 (Expected)

Master of Science in Computer Science and Technology

- **Focus Area:** Solar forecasting, spatio-temporal deep learning, and atmospheric data modeling.

SUPMTI (Higher School of Management, Telecom & Computer Science) — Oujda, Morocco — Sep 2021 – Jun 2024

State Engineering Degree in Computer Science & Information Systems (5-Year Master's Equivalent)

- **Focus Area:** Distributed Systems, Software Engineering, Clean Architecture, and Enterprise Cloud Systems.

ISTA Lazaret (Specialized Institute of Applied Technology) — Oujda, Morocco — Sep 2019 – Jun 2021

Specialized Technician Diploma in Software Development (TS)

- **Focus Area:** Object-Oriented Programming (OOP), Relational Databases (SQL Server/MySQL), Full-Stack Web Development.

FEATURED ENGINEERING PROJECTS

PV Forecast Benchmark Python PyTorch LightGBM Zenodo [Code | DOI: 10.5281/zenodo.23590205] — 2024 – Present

- **Standardized Benchmarking:** Multi-horizon benchmark evaluating solar generation across 50 operational sites, resolving accuracy tradeoffs.

Connect Check App Flutter Firebase Riverpod SQLite [GitHub] — 2024 – 2025

- **Reactive Mobile Engineering:** Real-time cross-platform application with live synchronization, Riverpod localization, and offline SQLite storage.

EcoBeach Environmental Platform ASP.NET Core 8 C# Clean Arch IoT — 2024

- **IoT Environmental Systems:** Clean Architecture backend monitoring coastal water quality, environmental sensors, alert thresholds, and reporting.

MystoGuess Mobile Game Flutter Dart Google Play [GitHub] — 2024

- **Published Game Application:** Mobile puzzle game on Google Play Store with dynamic animations, audio engine, and offline state persistence.

LANGUAGES

English: Fluent (Professional working & scientific research) | **French:** Proficient (Professional working proficiency) | **Chinese (Mandarin):** Conversational (HSK 3) | **Arabic:** Native | **Tamazight:** Native